Fiscal Year:	FY 2015	**Task Last Updated:**	FY 11/12/2014
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	NSCOR: Charged Particle Radiation and Resultant Oxidative Stress Elicit Deleterious Functional Changes in the Central Nervous System		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	grnelson@llu.edu	**Fax:**	FY 909-558-4035
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8364
Organization Name:	Loma Linda University		
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PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSCOR Space Radiation NNJ08ZSA003N
Start Date:	01/01/2010	**End Date:**	06/30/2016
No. of Post Docs:	15	**No. of PhD Degrees:**	
No. of PhD Candidates:	6	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 6/30/2016 per D. Risin/JSC, S. Monk/LaRC, and NSSC (Ed., 1/5/16) NOTE: Extended to 12/31/2015 per PI and NSSC information (Ed., 11/14/14)		
Key Personnel Changes/Previous PI:	FY2015 report: S. Rosi (U Cal San Francisco) and A. Allen (U Arkansas for Medical Sciences) sub-projects now established by separate sub-awards from prior association with J. Fike (U Cal San Francisco).		
COI Name (Institution):	Soltesz, Ivan (University of California, Irvine) Limoli, Charles (University of California, Irvine) Vlkolinsky, Roman (Loma Linda University) Raber, Jacob (Oregon Health & Science University) Rosi, Susanna (University of California San Francisco) Allen, Antino (University of Arkansas for Medical Sciences)		
Grant/Contract No.:	NNX10AD59G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed project will investigate responses of the mouse brain to several radiation exposure regimes corresponding to galactic cosmic rays and solar particle events. The project will characterize behavioral, electrophysiological, biochemical, and structural changes both in vivo and in vitro. Radiation-elicited oxidative stress mechanisms will be investigated as important mediators of the biological responses in both wild type and mutant animals. In addition to characterization of normal tissue responses, the potential enhancement of three late risks by radiation exposure will be explored. These are seizures, ischemic injury, and traumatic injury. Mathematical models of the neurotoxicity responses will developed for use in estimating risks and fitted to project-specific data and previously-published observations. The aims of the proposal will address HRP risk 1.2 and at least five of the specific gap questions identified by NASA: CNS-1, -3, -5, -6, and -11.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the responses of the central nervous system to charged particle radiation has two main benefits, risk assessment and optimization of radiotherapy. The general population and aviation crews are exposed to charged particles during cross country and trans-oceanic air travel at high altitudes as well as to radon in their homes and workplaces. While these exposures are not as high as those of astronauts they are not insignificant. Cancer risks have traditionally been the focus of health risk estimates but the new awareness of sensitive targets in the brain such as neurogenic centers in the hippocampus raises question about the negative effects of such radiation on late neurodegenerative diseases and potential decline of cognitive abilities. This project directly addresses cognitive and neurogenesis issues and indirectly addresses late neurodegenerative effects which share reactive oxygen chemistry with radiation as underlying mechanisms. Alzheimer's, Huntington's, and in particular, ALS diseases, are of interest. In the last ten years, the use of charged particles in radiation therapy has grown dramatically with proton and carbon ion treatment centers being established in the U.S., Europe, and Asia. Always limiting the effectiveness of radiotherapy is the response of normal tissue near the tumor site. The precise dose localization associated with charged particle therapy makes it ideal for tumors of the head and neck where careful avoidance of critical structures is critical. Knowledge of the dose tolerance of brain tissue and the reactions to relatively low doses is critical in optimizing these modalities of treatment.
Task Progress:	The CNS NSCOR project is proceeding under a no-cost extension but within budget incorporating recommendations of external review panels with adjustments to key personnel and contractual arrangements and plans for future experiments at Brookhaven National Laboratory and Loma Linda are underway. Cells and animals have been exposed to multiple species of charged particles as single acute exposures (H, O, Si, Ti, Fe) and one mixed ion exposure (H + Fe). Data to date show complex dose response relationships and dependence on radiation quality for all endpoints examined. Persistent elevated levels of oxidative stress are observed for murine and human neural precursor cells in vitro and brain regions in vivo following irradiation with a robust antioxidant response at 2 weeks post irradiation. Responses of the catalase overexpressing MCAT mouse strain are generally less impaired than in control animals. Different neuronal fields of the hippocampus (dentate gyrus, CA1) exhibit unique responses and there is evidence for continued evolution of brain tissue responses over the three months of observation. Radiation elicits cognitive, structural, and electrophysiological responses at doses below 0.1 Gy in complex and sometimes antagonistic patterns which don't necessarily correlate with neurogenesis or marker gene expression. Several lines of evidence suggest that inhibitory neurons are differentially more sensitive than excitatory neurons. Robust structural changes are seen for dendrite arbors and dendritic spines which are region and spine class specific. Novel innate immune responses have been observed after mixed ion exposure suggesting interactions between brain and peripheral immune compartments. Computer models have been improved for structural parameters and predict that subtle changes observed for intrinsic neuron properties post-irradiation can lead to altered network behavior in the hippocampus. Numerous joint manuscripts and presentations have been published by NSCOR team members and significant programmatic support has been provided by team members to the NASA HRP program.
Bibliography Type:	Description: (Last Updated: 03/26/2026)
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